// Julius Zhang

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github.com/JuI3s

Julius Zhang

Experience

Columbia University

Visiting Researcher Jun. 2026 - Present

- Conduct research in the Department of Mathematics, hosted by Andrew Blumberg.
- Computational optimal transport; arithmetic geometry applied to cryptography; security of AI-automated formal verification; homotopy theory.

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Researcher Apr. 2025 - Jul. 2026

- Research on new cryptographic constructions from lattice-based cryptography and commitment compression techniques. Manuscripts forthcoming.
- Building Jolt, open-source zero-knowledge virtual machine in Rust. Implemented and benchmarked torus compression, yielding a 3× reduction in commitment size; proved optimality with a conceptual reframing via Hilbert 90.Performance engineering across the proving stack, including sum-check optimization.
- Audited proof systems security: verify correctness of elliptic-curve optimizations and formalize folklore results into rigorous proofs.
- Evaluated lattice-based cryptography parameters against known attacks under various security regimes and performance constraints.

Millennium

Quantitative Researcher Jun. 2024 - Apr. 2025

- Convex optimization and robust statistical estimation for real-time trading systems.

Stellar Development Foundation

Consultant Jun. 2023 - Sept. 2023

- Byzantine fault tolerance, C++, stellar-core.
- Improved maximum transaction rate per second by 17% via changes to the consensus protocol, reducing number of communication rounds in overlay network. Formal verification in TLA+.

Distributed Systems and TCS Research, Stanford University

Researcher Sept. 2023 - Jun. 2024

- Under David Mazières, research in general open membership Byzantine fault tolerance protocols (Github), with applications to certificate authority management and automatic password recovery with hardware security modules.
- Research PCP under Aviad Rubinstein, reconstructed proof of the 2-to-2 Games Conjecture. Proved a result about Convex Continuous Class Complexity.

Topology Research, Stanford University

Researcher Jul. 2020 - May 2022

- Floer homotopy theory under Mohammed Abouzaid; related the complex cobordism Gysin map to the Grothendieck residue symbol and derived a recursive formula for Gromov–Witten quantum product corrections.
- Research under Ciprian Manolescu. Floer theory, Seiberg-Witten, knot invariants. Produced new potential counterexamples for the smooth Poincare conjecture.

Selected Papers and Manuscripts

- Zachary DeStefano, Noah Golub, Zile Huang, **Julius Zhang**, Sam Frank, and Michael Walfish
  Spain: Succinct Proofs for Numerical Computations. [PDF]
  USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2026.
- Julius Zhang.
  The Relative Trace-Zero Subgroup of the Barreto-Naehrig Curves. [PDF]
  Cryptology ePrint Archive, 2026.
- Julius Zhang.
  Complex Cobordism and Formal Group Law. [PDF]
  Stanford University, 2024.

Talks

- *p-Ordinary Cohomology for* $\mathrm{SL}(2)$ *over Number Fields*.
  Hida Theory Seminar, Columbia University, 2026.

- *Construction of Seiberg-Witten-Floer stable homotopy types*.
  Homotopy Theory and Manifolds Seminar, Columbia University, 2026.
- *Bordered Heegaard Floer Homology and the Surgery Exact Triangle*.
  Stanford University, 2022.
- *Knots, links, and the search for exotic 4-spheres*.
  Stanford University, 2021.

Education

Courant Institute of Mathematical Sciences, New York University

Ph.D. in Computer Science June 2026 - Present

Stanford University

M.S. in Computer Science; B.A. in Mathematics Sept. 2019 - Jun. 2024
General Wang Yaowu Scholarship.

Skills

Computer Science. Rust, C/C++, Python; optimization and robust estimation; high-performance numerical systems; cryptography and verifiable computation; formal verification.
Mathematics. Number theory, algebraic geometry, homotopy theory.